

Microprocessor-Based Robust Digital Control for UPS with Three-Phase PWM Inverter

Youchi Ito and Shoichi Kawauchi

Abstract— This paper describes a novel method of robust (insensitive to system parameter variations and load current changes) and fast digital control for uninterruptible power supply (UPS) with a three-phase PWM inverter. The purpose of this paper is to propose a method by which characteristics better than those by conventional methods are obtained using an algorithm simpler than that of conventional methods. The experiments show that the purpose is achieved and the proposed method offers a total harmonic distortion of 0.62% of the output voltage waveform at a full nonlinear load. The analysis shows stability of the method is sufficient. Three features of the method are; (a) A capacitor current observer for stabilization and a disturbance observer for robustness are used to compensate the time lag by the computation and the disturbances, in a minor loop of the capacitor current through an inductor-capacitor filter of the inverter. (b) New models of the inverter and the disturbances are established to simplify these observers. (c) The output voltage control loop can be designed easily and exactly because the minor loop realizes a rapid and robust control of the current.

I. INTRODUCTION

WITH the increasing use of computers and factory automation systems, higher reliability of uninterruptible power supply (UPS) in these systems is required. UPS control circuits must be digitalized with microprocessors including a digital signal processor (DSP), because digital circuits essentially have no aging and high reliability, and are easier to implement than analog circuits.

It is demanded that PWM inverters in the UPS have output voltage waveform with low total harmonic distortion (THD) and little output voltage variation even with nonlinear loads. To obtain such characteristics, disturbances caused by the UPS load currents, phase coupling components, and plant parameter variations must be compensated satisfactorily. The characteristics of digitally controlled inverters depend on the sampling time of control processors and the time lag by the computational time of the control processors. A basic condition to improve the performances is to make the time lag and a sampling period shorter as far as possible. Hence, the essential requirements for digital control methods are complete cancelation of the disturbance, compensation of the time lag, and a simple control algorithm. Even if control processors are progressed, these requirements remain unchanged.

This paper proposes a digital control method which satisfies these requirements better than the conventional methods [1] to [4]. To obtain a fast and robust control of the output voltage,

it is necessary to realize a fast and robust control of the capacitor current through the inductor-capacitor (L - C) filter of the inverter, because the UPS output voltage is the voltage across the capacitor in the L - C filter. From this, the capacitor current minor control loop is adopted in the output voltage major control loop, and the proposed method uses a current and disturbance observers in the minor loop.

Three features of this minor loop are following: a) To obtain a fast response current control, the current observer predicts the capacitor current to compensate the time lag, because design of a dead beat compensator to provide quick control demands the compensation. This observer stabilizes the control loop; b) To get a robust current control, the disturbance observer estimates a value which contains all disturbances such as load current and system parameter variations, because all the disturbances must be canceled. This observer provides robustness; c) To realize simple control algorithm, an essential model of the inverter main circuit (plant) and a novel disturbance model are produced, and the observers are derived from these models. The disturbance observer is of the structure quite different from disturbance observers [4], [5], and [7] for a robust controls. In addition, the major loop can adopt a proportional compensator because of the fast and robust minor loop.

Theoretical analysis and experiments show the validity of the proposed method. The analysis reveal that the method is stable enough to a plant parameter variation. For a 10 kVA experimental inverter, when the method is implemented using a TMS320E15 (DSP) as a digital controller, the current sampling frequency of 36 kHz as high as two times the PWM carrier frequency can be obtained. This means that the proposed control algorithm is about one half that of the conventional methods. Experimental results show that the proposed method produced an output voltage with THD less than 1%. This value of THD is about one third those given by the conventional methods.

II. MODELING OF INVERTER AND BASIC VOLTAGE CONTROL SYSTEM

In this section, a single-phase model of the inverter is derived. Then, an output voltage control system using the model is described. A discrete-time capacitor current control system is obtained.

A. Main Circuit Model

Fig. 1(a) shows the inverter main circuit which consists of L - C filters in the inverter output, IGBT's as switching devices,

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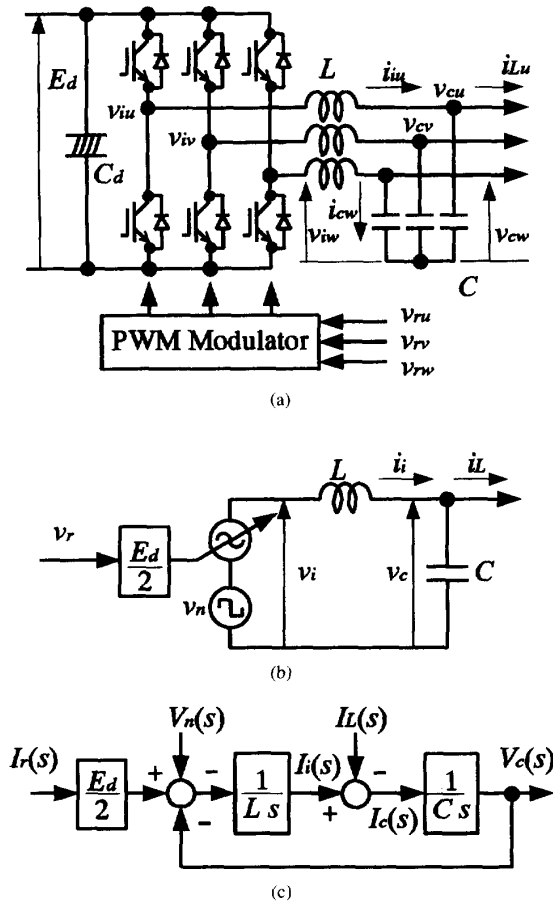


Fig. 1. Inverter main circuit model (a) Three-phase PWM inverter for UPS; (b) Equivalent single-phase circuit; (c) Main circuit (plant) model.

and a dc capacitor filter in the inverter input. v_{ru} , v_{rv} , and v_{rw} are modulation signals which are the input to the PWM modulator in the figure. v_{iu} , v_{iv} and v_{iw} are the inverter output with respect to the filter capacitor neutral point.

$$\begin{aligned} v_{iu} &= \frac{E_d}{2} v_{ru} + v_{hu} - v_N \\ v_{iv} &= \frac{E_d}{2} v_{rv} + v_{hv} - v_N \quad v_{iw} = \frac{E_d}{2} v_{rw} + v_{hw} - v_N \quad (1) \\ v_N &= (v_{hu} + v_{hv} + v_{hw})/3 \end{aligned}$$

where v_{hu} , v_{hv} , v_{hw}

: Each-phase IGBT on-state voltage

E_d : Inverter dc voltage

The inverter phase voltages interfere with each other because v_N fluctuates depending on IGBT switching conditions. v_N is a phase coupling voltage. An equivalent single-phase circuit of the inverter shown in Fig. 1(a) is presented in Fig. 1(b). In this figure, the variables are described without subscript u , v , or w (for example, v_{ru} , v_{rv} , or v_{rw} is v_r), because the inverter is a symmetrical circuit. And, v_h minus v_N is v_n .

Equations for this circuit are

$$L \frac{di_i}{dt} = v_i - v_c C \frac{dv_c}{dt} = i_i - i_L \quad (2)$$

where i_i : Inverter (filter inductor) current

v_i : Inverter voltage defined by (1)

v_c : Output (filter capacitor) voltage

i_L : Load current

From (2), a block diagram of the single-phase main circuit model as shown in Fig. 1(c) is obtained. The diagram is in the continuous time domain. It shows that the coupling voltage v_n and the load current i_L are disturbances.

B. Output Voltage Control System

The basic control system of the output voltage (the filter capacitor voltage) as shown in Fig. 2 is constructed using the main circuit model in Fig. 1(c). $D(z)$ and $E(z)$ are the compensators in the control system. The time lag T_d is described as $\exp(-T_d s)$. The PWM inverter is expressed by a zero-order hold, a sampler and a proportional gain $K_a = E_d/2$. This voltage control system has a filter capacitor current feedback path (the current control system) as a minor control loop. This loop is essential to increase stability of the voltage control system and to control the capacitor current exactly. The capacitor voltage feedforward path in the minor loop reduces a disturbance.

When this system is implemented with TMS320E15 (DSP) as control processor, the proposed control algorithm can offer the current-sampling frequency f_{si} is two times as much as the PWM frequency which is 18 kHz. According to Shannon's theorem, f_{si} must be at least two times as high as the PWM carrier frequency to sample the current which contains components of the switching harmonic voltage v_n . If the sampling frequency is high, the high natural frequency of the system can be designed and the current control performance is improved. However, upper limit of the sampling frequency depends on the control algorithm and the processor. Thus, the f_{si} is given 36 kHz.

The voltage-sampling frequency f_{se} is 9 kHz which is a quarter of the f_{si} . If the current follows its command by the minor loop, a natural frequency of 1 kHz for the voltage control system may be sufficient for a system frequency of 50 or 60 Hz because the voltage control system need not compensate the disturbances. The voltage control system can be designed in the continuous time domain because generally sampling and time lag influences on the control performances can be neglected in the system if the f_{se} is about 10 times as high as the natural frequency.

C. Current Control System in Discrete Time Domain

To obtain high performance of the current control, the minor loop in Fig. 2 must be discussed in the discrete time domain. If the minor loop is designed to provide that the current is controlled as quickly as possible and the disturbances are canceled as fast as possible, it is necessary to consider the sampling and the time lag influences. The current control

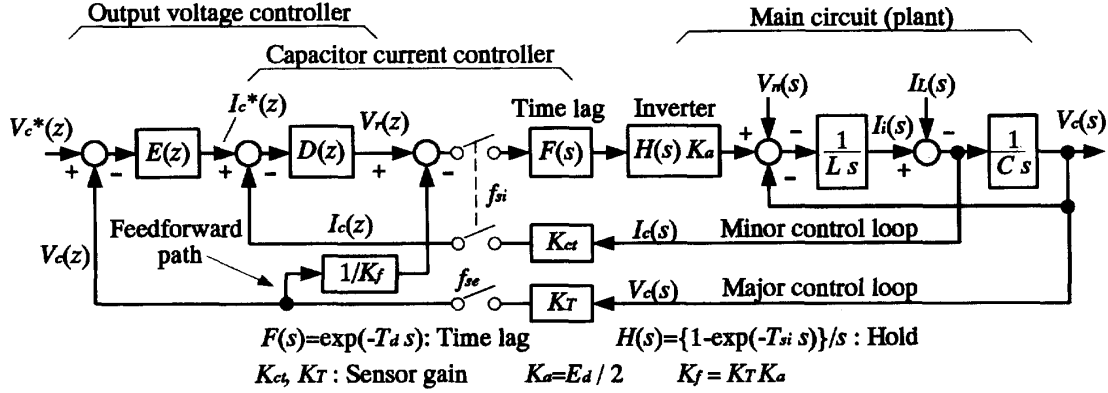


Fig. 2. Basic output voltage control system.

loop is transformed into the domain by the modified z-transformation to treat the time lag accurately. The transformed control system is shown in Fig. 3(a). The sampling period for each variable is defined as shown in Fig. 3(b). The sample is represented by (n) . The next sample is denoted by $(n+1)$. The sample between (n) and $(n+1)$ is given by (n, m) . The m is the time-lag coefficient. The time lag T_d is designed $(1-m)T_{si}$. When T_d is zero, $m = 1$. When T_d is T_{si} , $m = 0$. The voltage $v_r(n)$ is calculated using $i_c(n-1, m)$ and the time lag T_d .

III. TIME LAG AND DISTURBANCE COMPENSATION BY OBSERVERS

In this section, the proposed current control system with the observers is described in the discrete time domain. Firstly, the time lag in the minor loop is compensated for stabilization. The compensation is performed by using a capacitor current observer based on the current control system. Next, the disturbances are compensated by using a disturbance observer for robustness after the time lag is compensated.

A. Time Lag Compensation by Capacitor Current Observer

The time lag in the minor loop is mainly caused by computational time of the DSP in the current control. The characteristics of digital controllers are inferior to those of analog ones, because natural frequencies of digital control systems must be designed to be lower than those of analog control systems, if the time lag exists.

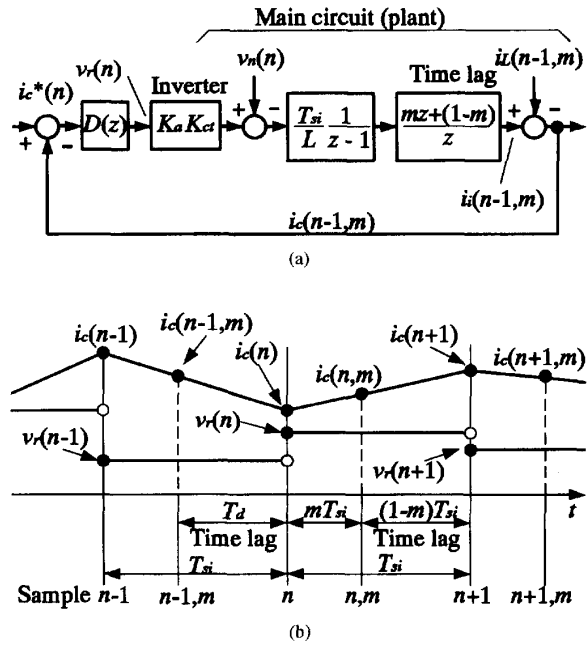


Fig. 3. Discrete-time current control system (a) Capacitor current control system; (b) Definition of sampling period.

The time lag can be compensated by feedback of capacitor current $i_c(n)$ instead of the current $i_c(n-1, m)$. The $i_c(n)$

$$i_c(n) = i_c(n-1) + T_{si} K_b v_r(n-1) - \frac{T_{si}}{L} v_D(n-1) \quad (3.1)$$

$$i_c(n-1, m) = i_c(n-1) + m T_{si} K_b v_r(n-1) - \frac{T_{si}}{L} v_D(n-1, m) \quad (3.2)$$

Where $K_b = K_{ct} \frac{E_d}{L}$: Plant parameter

$v_D(n-1) = v_n(n-1) - \frac{L}{T_{si}} \{i_L(n) - i_L(n-1, m)\}$: Disturbance

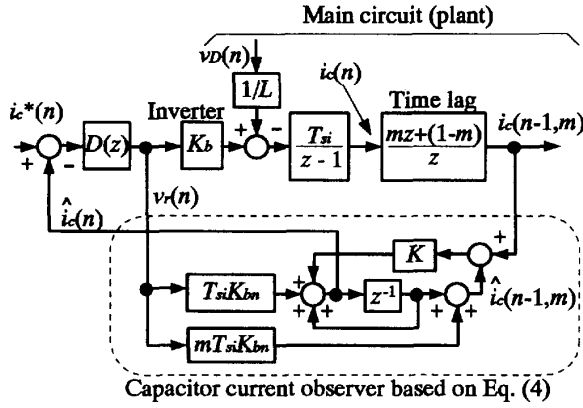


Fig. 4. Time-lag compensation using observer.

is predicted by the observer. The observer is derived from the state equation of the plant [6]. The state equation of the plant in Fig. 3(b) is (see (3.1) and (3.2) at the bottom of previous page)

When the disturbances, i_L and v_n are zero in the above equation, the current observer gives the predicted value $i_c(n)$ is

$$\begin{aligned} \hat{i}_c(n) &= \hat{i}_c(n-1) + T_{si}K_{bn}v_r(n-1) \\ &+ K \{ i_c(n-1, m) - \hat{i}_c(n-1) - mT_{si}K_{bn}v_r(n-1) \} \end{aligned} \quad (4)$$

Where K : Observer gain

K_{bn} : Plant parameter nominal value

Fig. 4 shows the current control system in which the time lag is compensated by using this current observer. The settling time of current estimates is designed with gain K because the observer characteristic equation is described as $z + 1 - K = 0$. When $K = 1$, the dead-beat observer is realized.

B. Design of Compensator in Capacitor Current System

A dead-beat current controller is realized when the compensator $D(z)$ is equal to $1/(K_{bn}T_{si})$ from the characteristic equation of the current system where the time lag is compensated; $z - 1 + T_{si}K_{bn}D(z) = 0$. However, this current control system does not provide accurate control because the observer does not predict the n sample current exactly, except when $v_n = 0$, $i_L = 0$ and the plant parameter values are nominal values.

C. Disturbance Compensation by Disturbance Observer

To provide exact current control, the disturbances in the system must be compensated. The disturbance observer carries out the compensation. An equivalent block diagram of the current control system is shown in Fig. 5(a). i_D is assumed to be a variable contains all disturbances such as v_n , i_L and

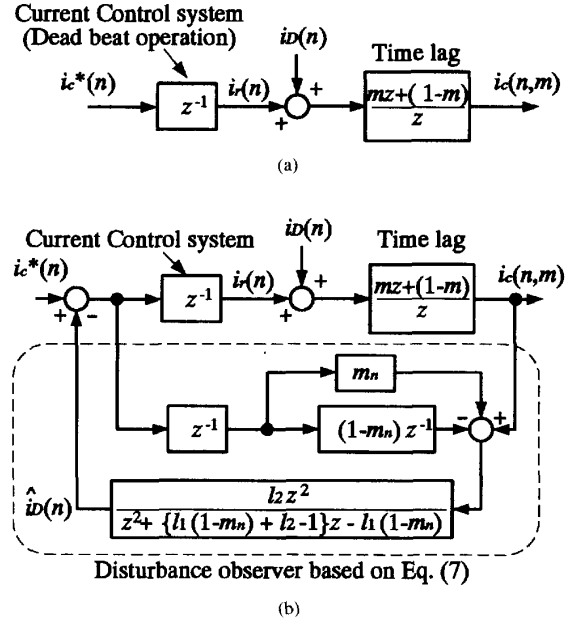


Fig. 5. Disturbance compensation using observer (a) Equivalent block diagram of current control system in Fig. 4; (b) Disturbance observer block diagram.

parameter variations. i_D is compensated by the feedforward path of i_D which is estimated by the observer.

The disturbance observer is derived by the plant state equation in the same manner in the case of the current observer. An observer plant model is the system shown in Fig. 5(a). This system is defined as a second order system where the state variables are $i_r(n)$ and $i_D(n)$, and the output variable is $i_c(n, m)$. i_D is assumed to be $i_D(n) = i_D(n-1)$, in the steady state. The state equation of this system is

$$\mathbf{x}(n) = \mathbf{A}\mathbf{x}(n-1) + \mathbf{b}i^*(n-1) \quad (5.1)$$

$$i_c(n, m) = \mathbf{c}\mathbf{A}(m)\mathbf{x}(n-1) + \mathbf{c}\mathbf{b}(m)i^*(n-1) \quad (5.2)$$

$$\text{where } \mathbf{x}(n) = (i_r(n), i_D(n))^T$$

$$\mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \mathbf{c} = (1, \quad 1)$$

$$\mathbf{A}(m) = \begin{pmatrix} 1-m & 1 \\ 0 & 0 \end{pmatrix} \quad \mathbf{b}(m) = \begin{pmatrix} m \\ 0 \end{pmatrix}$$

The observer is given by

$$\begin{aligned} \hat{\mathbf{x}}(n) &= \mathbf{A}\hat{\mathbf{x}}(n-1) + \mathbf{b}i^*(n-1) \\ &+ \mathbf{l} \{ i_c(n, m) - \mathbf{c}\mathbf{A}(m)\hat{\mathbf{x}}(n-1) + \mathbf{c}\mathbf{b}(m)i^*(n-1) \} \end{aligned} \quad (6)$$

where

$$\hat{\mathbf{x}}(n) = (\hat{i}_r(n), \hat{i}_D(n))^T : \text{Estimated vector}$$

$$\mathbf{l} = (l_1, \quad l_2) : \text{Observer gain vector}$$

$$\mathbf{A}_n(m) = \begin{pmatrix} 1-m_n & 1 \\ 0 & 0 \end{pmatrix} \quad \mathbf{b}_n(m) = \begin{pmatrix} m_n \\ 0 \end{pmatrix}$$

$$m_n : \text{Time-lag coefficient in observer}$$

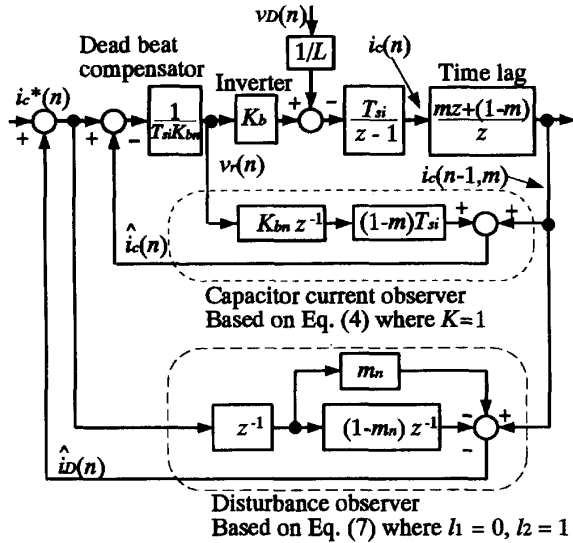


Fig. 6. Proposed current control system with two types of dead beat observers and dead beat compensator.

Solving (6) for $i_D(n)$,

$$\hat{i}_D(n) = \frac{l_2 z^2 [i_c(n, m) - \{(1 - m_n)z^{-2} + m_n z^{-1}\} i_c^*(n)]}{z^2 + \{l_1(1 - m_n) + l_2 - 1\}z - l_1(1 - m_n)} \quad (7)$$

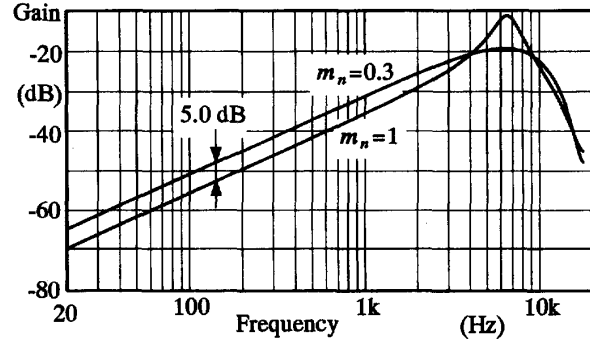
The disturbance i_D is compensated by using this observer shown in Fig. 5(b). The observer characteristic equation is the same as the denominator of (7). The dead-beat observer is given when $l_1 = 0$ and $l_2 = 1$.

D. Proposed Current Control System

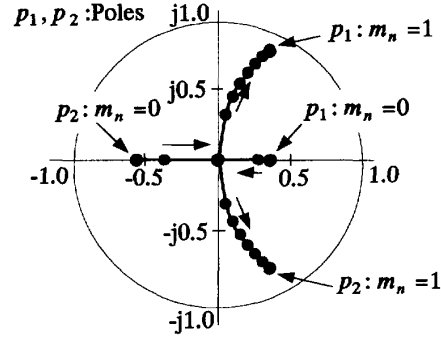
In Fig. 6, the proposed current control system with two dead-beat observers and a dead-beat compensator is shown. These observers are the capacitor current observer with $K = 1$ and the disturbance observer with $l_1 = 0$ and $l_2 = 1$. The current observer contributes to the system stability. The disturbance observer helps to make the system robust. The compensator employs the proportional gain, $1/K_b T_{si}$. The current control system is very simple. Since this system is realized with a simple control algorithm, high current-sampling frequency is achieved.

IV. ANALYSIS OF CONTROL SYSTEM

Stability of the proposed system must be confirmed. First, theoretical disturbance suppression characteristics of the proposed current control system are considered. Analytical results show that the suppression characteristics by the observer with $m_n = 1 \neq m$ are better than those by the observer under $m_n = m$, although time lag exists. Stability of this system for parameter variations is examined next. Lastly a design method for the output voltage control system is described.



(a)



(b)

Fig. 7. Relation between disturbance suppression and time lag coefficient in disturbance observer. (a) Disturbance suppression characteristics of current control system where $m = 0.3$. (b) Root locus in current control system where $m = 0.3$.

A. Relation Between Disturbance Suppression and Time-lag Coefficient

The suppression characteristic is given by frequency response curves of i_c to i_D for Fig. 5(b). The curves are drawn by putting $z = \exp(-T_{si}s)$. The time-lag coefficient m of the experimental controller is 0.3 which is explained in the next section. The curves are shown in Fig. 7(a) for the cases of $m_n = 1$ or $m_n = 0.3$ with $m = 0.3$. Fig. 7(b) shows a root locus obtained when m_n changes from 0 to 1 with $m = 0.3$. If $m_n = 0.3 (= m)$, all poles are located at the origin. As m_n increases, two poles move to the outside of the unit circle. Therefore, when the observer employs $m_n = 1$, the system does not provide a dead-beat control. The suppression characteristic is improved by 5.0 dB at 2 kHz. The damping ratio of the system is 0.2 (the system is stable).

B. Stability of Current Control System with Parameter Variations

A root locus of the proposed system is shown in Fig. 8 to investigate stability of the system when the plant parameters change. Fig. 8(a) shows the root locus for $m = m_n = 0.3$. Fig. 8(b) shows it for $m_n = 1$ and $m = 0.3$. The parameter, K_b , is changed from -100% to 100% . The system is stable up to

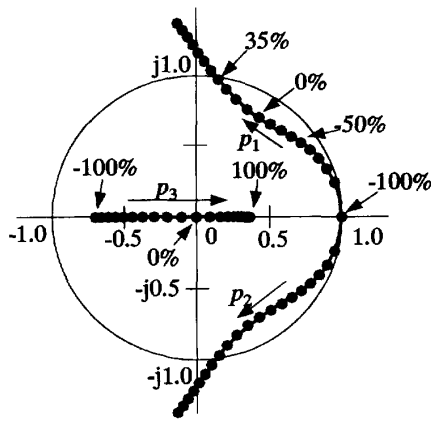
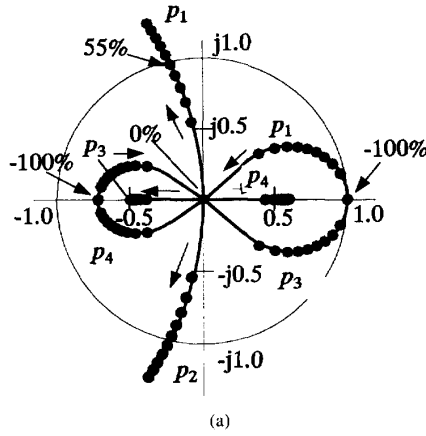


Fig. 8. Root locus of current control system with plant parameter variation. (a) $m_n = 0.3$, $m = 0.3$. (b) $m_n = 1$, $m = 0.3$.

a variation of 55% if m_n is 0.3, and up to 35% if m_n is 1. On the other hand, the system is stable to a variation of -100% for both cases. These parameter variations with stability allow the application of this system to the UPS.

C. Design of Output Voltage Control System

The transfer function of the current control system is approximated by the proportional gain, $1/K_{ct}$ because the current control system natural frequency can be much higher than the voltage control system natural frequency. The output voltage control system is shown in Fig. 9 in the continuous-time domain. This system does not contain any disturbances because the current control system suppressed them and regulates the current completely. When the compensator $E(z)$ is designed, only the command response is considered. The proportional compensator K_{pe} is adopted for $E(z)$. The characteristic equation of the system in Fig. 9 is $s + k_{pe}K_T/K_{ct}C = 0$. K_{pe} was chosen to produce a natural frequency of 1 kHz.

This control system has a phase delay of 2.86° and a steady-state error of 0.12% when the command frequency is 50 Hz, which is the power source frequency. When this system is applied for a UPS, the phase delay and the steady-state error

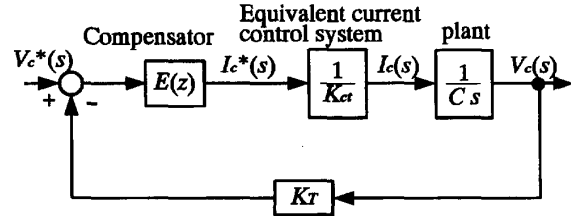


Fig. 9. Construction of output voltage control system.

can be compensated by modifying the command corresponding to the above characteristics, because the characteristics change little.

V. EXPERIMENTAL RESULTS

A. Construction of Experimental Main Circuit and DSP Controller

Fig. 10(a) shows the construction of a 10 kVA experimental main circuit and a Digital controller based on the systems in Fig. 6 and Fig. 9. The computational time of the two-phase current control is $19.4 \mu s$, when the DSP TMS320E15 is employed. Then, the time-lag coefficient m in the observers is 0.3 according to the definition in Fig. 3(b). The current sampling frequency is 36 kHz. It is two times as much as the PWM carrier frequency. The output voltage sampling frequency is 9 kHz. The DSP samples and controls the phase-U and phase-W capacitor currents and the phase-U and phase-W output voltages. It outputs the modulation signals v_{ru} , v_{rv} and v_{rw} as control variables to the PWM modulator.

The PWM modulator converts each-phase modulation signal into the IGBT triggering signals for each phase during every sampling period. The modulator consists mainly of timer IC's. The virtual carrier frequency is 18 kHz although the modulator does not need a real carrier signal. Fig. 10(b) shows a timing chart for all of the signals.

B. Experimental Results of Output Voltage Characteristics

Fig. 11 shows the output line voltage and the load current waveforms with a 10 kVA capacitor-input rectifier. In Fig. 11(a) where m_n is 0.3, the THD is 1.06%. In Fig. 11(b) where m_n is 1, the THD is 0.62%. The THD is improved by 4.7 dB. This value is similar to the analysis result of 5.0 dB. It is shown that the THD depends on the disturbance suppression characteristic of the current control system.

Steady-state characteristics of the output line voltage are shown in Fig. 12(a) and (b). When the dc voltage is changed from 440 V (nominal) to 340 V (-24%), the THD is increased by 1.5 dB with the rectifier load. The output voltage regulation is -0.15% with no load in Fig. 12(a). The regulation is 0.3% for the entire load range in Fig. 12(b).

VI. CONCLUSIONS

A fast response and robust digital control method for the inverter in UPS was presented. The performance of two types

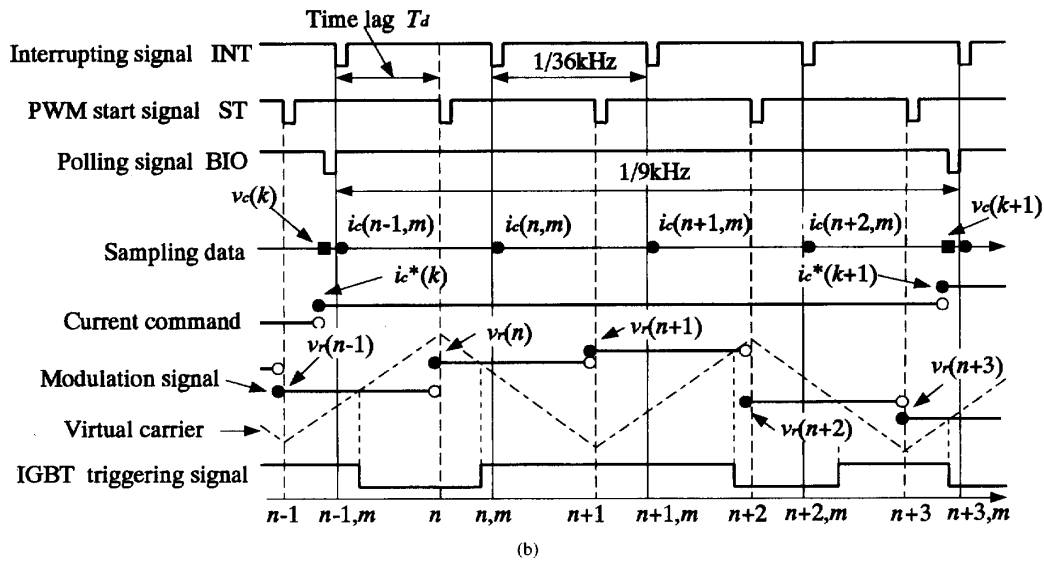
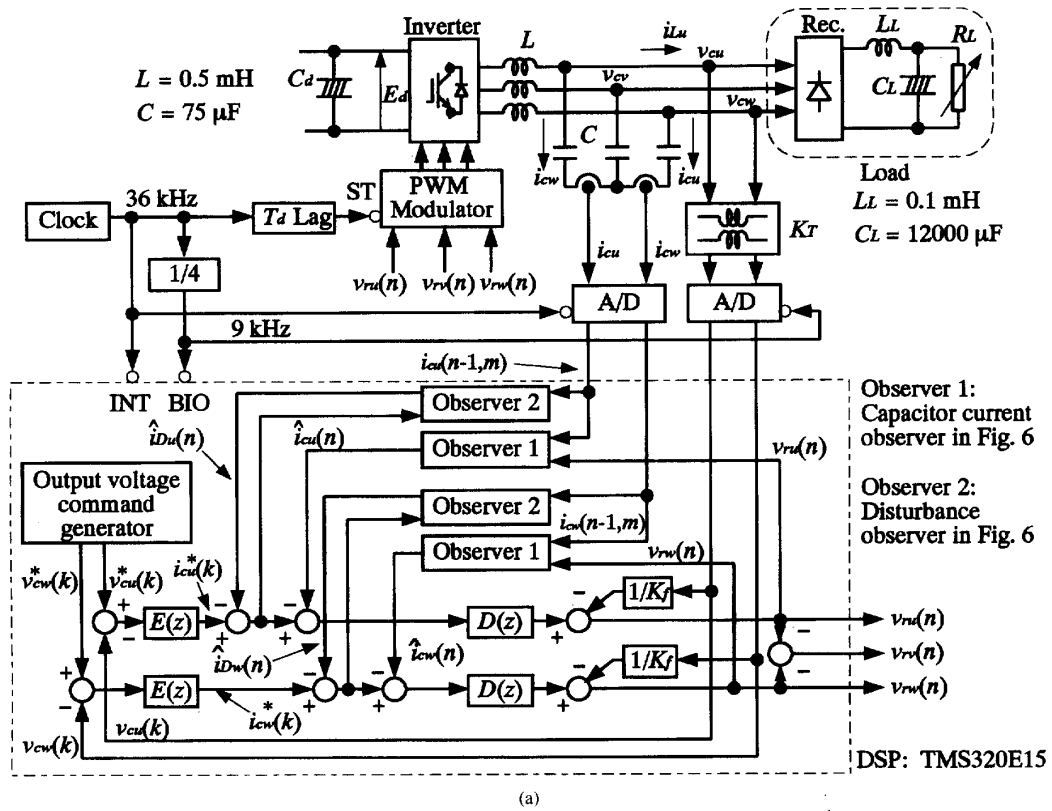


Fig. 10. Block diagram of 10 kVA experimental inverter for UPS (a) Main circuit and DSP controller; (b) Timing chart of DSP controller.

of the observers and the high sampling frequency obtained by the simple algorithm allowed the proposed method to achieve quick and exact current control and to provide low THD of the output voltage waveform. The experimental results demonstrated that the proposed control algorithm

is about one half those of the conventional methods and the value of THD is about one third those given by the conventional methods. The analysis showed the stability of the control system is satisfactory on application to the UPS.

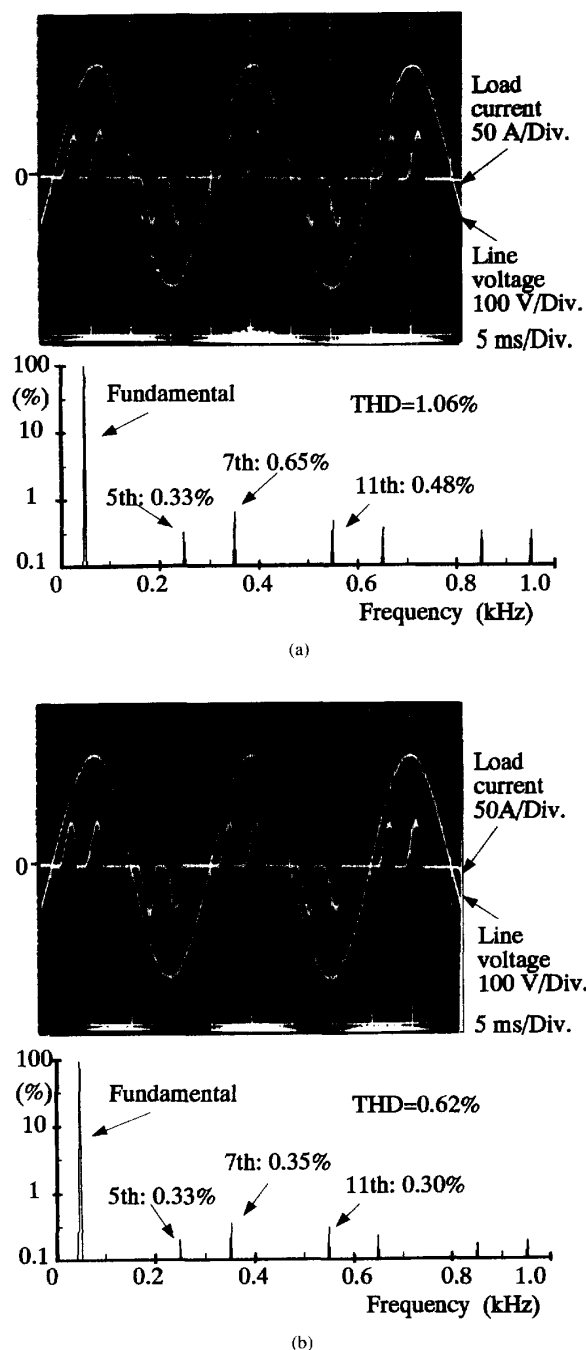


Fig. 11. Output line voltage (Phase U-W) and load current (phase U) waveforms under 10 kVA capacitor-input rectifier load (a) $m_n = 0.3$, $m = 0.3$; (b) $m_n = 1$, $m = 0.3$.

When the method is applied to UPS products, the use of recent multi-purpose microprocessors better than TMS320E15 in functions, cost and calculation time must improve the cost performance of the products. The method can achieve full digitalization of small-capacity UPS, although only large-

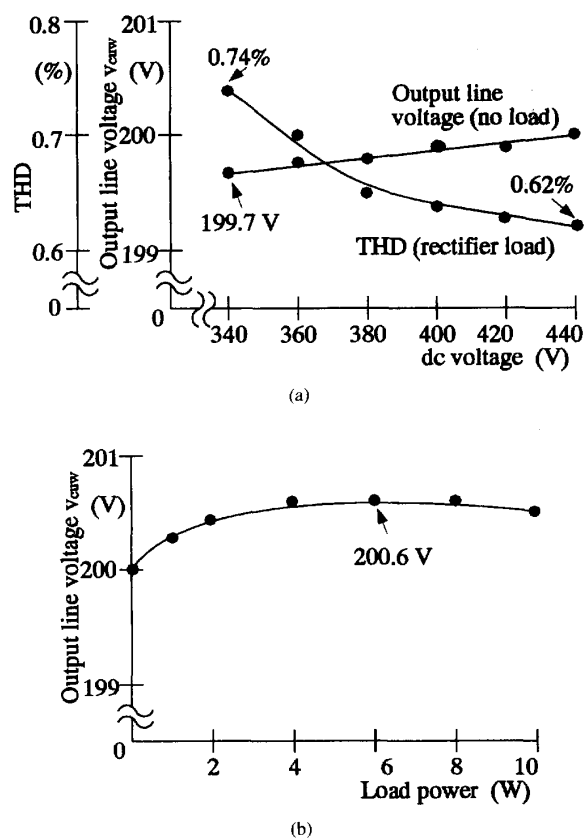


Fig. 12. Steady-state characteristics of output line voltage (a) To dc voltage; (b) To load power.

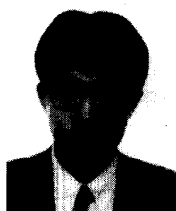
capacity UPS have been fully digitalized. It can be applied to a wide range of products which need current controls.

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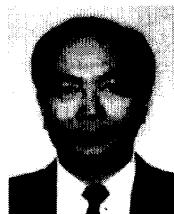
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